Screening the Firm Resource:

De-rating Calibration, Adverse Selection and Lock-in in Spain's 2026/2027 Capacity Mechanism (SA.112483) — an admissible-space audit with hourly ELCC data (2019–2024)

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Abstract

A capacity auction is a *screening* mechanism: the de-rating factor and the critical-hour window form a "slot" that makes each technology reveal —and be paid for— its firmness type. We audit the slot of Spain's 2026/2027 capacity mechanism (9 € billion over ten years, approved as State aid SA.112483 (European Commission, 2026)). Crucially, Spain's national de-rating coefficient (PO Section 7.1) —the average, over the 5–10% lowest-coverage-margin hours, of firm power over installed capacity— is methodologically the Effective Load Carrying Capability (ELCC) we estimate from hourly ENTSO-E data (BZN|ES, 2019–2024) and is identical to ERAA method 12(7)(c): the realised ELCC *is* essentially the regulatory coefficient, so the slot we audit is the real one. We show it mis-screens firmness as four facets of one *adverse-selection* failure: aggregated hydro under-credits dispatchable reservoir (ELCC ≈0.73, measured by mean production rather than available power); an imported window credits solar at ≈0.03 where the Spanish bimodal profile gives ≈0.13; the gas coefficient is mis-specified under the regulation's own rule (declared availability → fuel-blind, or production → contaminated by the 2022 Iberian Exception, +0.072 on production); and stress is correctly defined on physical coverage margin, not price (MIBEL Jaccard ≤0.10). The headline distortion is timing: the first auctions (Q1 2027) run the national slot while the standardised EU de-rating factors reach full coverage only with the ERAA 2029 edition, and new-investment contracts run up to fifteen years —so a mis-screened storage or solar coefficient is locked in a decade before the cleaner EU benchmark lands, precisely where the investment signal lives. The national coefficient is moreover static, blind to the penetration- and mix-dependence of firmness that ACER itself now recognises (Decision 04/2026). Priced against the value of lost load (22,879 €/MWh; reliability standard 1.5 h LOLE/year), these mis-priced firm MW misallocate a multi-year, ≈900 € million/year investment stream.

Keywords: Capacity mechanism, SA.112483, screening, adverse selection, de-rating factors, ELCC, *missing money*, lock-in, regulatory arbitrage, ENTSO-E, MIBEL.

1 Introduction

1.1 Motivation: a 9 € billion mechanism that pays per firm megawatt

The Spanish capacity mechanism has moved from a regulatory proposal to a budgeted transfer. Its enabling provision had a turbulent passage —introduced by Royal Decree-law 7/2025 (Jefatura del Estado, 2025), not ratified and repealed by the Congreso on 22 July 2025, then reinstated as Article 33(13) of the Electricity Sector Law (Ley 24/2013) by the eighth final provision of Royal Decree-law 7/2026 (Jefatura del Estado, 2026), which is the live legal basis. On 29 May 2026 the European Commission authorised the scheme under State aid rules: a *market-wide* mechanism of 9 € billion over ten years (~900 € million/year) in which the transmission system operator (TSO) "remunerates all the capacity needed to meet the reliability standard" and beneficiaries "compete based on the amount of aid requested per MW of capacity available during a scarcity event" (European Commission, 2026). Approval rests on Article 107(3)(c) TFEU and the CEEAG 2022 Guidelines (European Commission, 2022), and declares alignment with most of the best practices of the *Clean Industrial State Aid Framework* (CISAF) (European Commission, 2025); the first auction is expected in Q1 2027. The detailed ministerial order remains a *draft* notified under SA.112483, so the operational parameters we audit (auction horizons, contract durations, the delegation of firmness coefficients to the TSO) derive from that draft or from EU law, not from settled national primary law —a status we flag throughout. That a repealed-then- reinstated enabling provision and an unpublished operational order govern a 9 € billion, multi-year commitment is itself part of the path-dependence risk this paper identifies.

The operative phrase —"per MW available during a scarcity event"— is exactly the definition of ELCC. The *derating factor* is the fraction of nominal capacity each technology can accredit as firm before the auction; multiplied by installed capacity and by the clearing price, it sets each project's capacity revenue. Mis-calibrating it is not an academic rounding error: it mis-allocates 900 € million a year and emits a biased investment signal over a ten-year horizon.

1.2 A screening lens

A capacity auction is, in effect, a screening device. The *derating factor* and the *critical-hour window* together fix the implied price per firm MW, and hence which technologies find it worthwhile to participate. A calibration that over-credits a low-firmness resource and under-credits a high-firmness one therefore induces *adverse selection*: the low type enters and is over-paid, the high type is under-paid and crowded out. Three implications follow. First, the de-rating method is the channel through which each technology's firmness is inferred from data, so the window, the production-versus-availability measure and the historical period are calibration choices that can mis-reveal a type. Second, because the auction sets a relative price across technologies, a mis-priced firm MW is a distorted signal; and because new-investment contracts run up to fifteen years, the distortion misallocates a decade-long investment stream. Third, the relevant standard is the Commission's own: a capacity mechanism must not (i) raise consumer prices, (ii) grant undue advantages to specific operators, nor (iii) hinder cross-border flows (European Commission, 2026). The remainder of the paper shows that Spain's national coefficient mis-screens firmness along four facets of a single adverse-selection problem (§6.1–§6.4), that the mis-pricing is locked in before the EU benchmark matures (§7–§8), and prices the distortion against the value of lost load.

1.3 Research question

What are the empirically justified *derating factors* for the relevant technologies (nuclear, wind, solar, reservoir hydro, run-of-river, pumped storage, gas) in the peninsular system 2019–2024, and what auction distortions does the 2026/2027 CM generate if calibrated with the usual shortcuts (aggregated hydro, imported benchmarks, uncleaned history, price-based activation)? To what extent are the findings robust to the temporal granularity of the data and to the internal split of hydro capacity?

1.4 Contribution

1. **ELCC from real hourly data** (ENTSO-E 16.1.B&C, BZN|ES, 2019–2024) instead of token-free daily generation (REData). For dispatchable technologies with a concentrated temporal profile, the hourly peak capacity factor differs materially from the daily one.

2. **Disaggregation of hydro** into its three sub-technologies (reservoir, run-of-river, pumped storage). The traditional aggregate conceals an adequacy heterogeneity of factor 5×, validated through a sensitivity analysis over five capacity-split scenarios.

3. **Quantification of the Iberian-Exception bias** on the *production* channel (+0.072 on gas dispatch, 2022–2023) and a structural diagnosis of its differential transmission: it biases technologies the regulation screens by production (hydro, solar, wind) directly, while for gas —screened by declared availability— the distortion is a table-versus-rule mis-specification, not a contamination of the availability series.

4. **Translation into design risks**: we map each statistical finding onto a named auction risk of the 2026/2027 CM (adverse selection, *missing money*, regulatory artifact, price-based activation), with the corresponding calibration recommendation.

2 Literature

Garver (1966) introduced ELCC as a metric of equivalence between capacity additions and demand reductions: a unit with ELCC=1.0 is equivalent, from an adequacy standpoint, to a permanent 1 MW reduction of peak demand. It is this equivalence that the 2026/2027 CM monetises by paying "per MW available in scarcity" (European Commission, 2026).

Mills and Wiser (2013) and Madaeni et al. (2013) formalise the various approaches to computing ELCC under growing renewable penetration, identifying the "saturation" or dilution effect of marginal ELCC as the installed capacity of technologies with correlated profiles increases.

The debate over the need for capacity mechanisms proper —versus an *energy-only* design— is consolidated in Joskow (2008) and Cramton et al. (2013). Batlle and Pérez-Arriaga (2008) discuss specific design criteria for liberalised markets, prefiguring elements of the Spanish design. The current European framework —Regulation (EU) 2019/943 (European Union, 2019), CEEAG 2022 (European Commission, 2022) and CISAF (European Commission, 2025)— requires the CM to be proportionate and non-discriminatory, which in practice hinges on the quality of the *derating factors*.

The European benchmarks published by National Grid ESO (2024), EirGrid and SONI (2023) and PSE (2023) serve as comparative references, but their transferability to the Spanish case is limited by differences in climatic profile and technology mix —a point that Axis 2 (Section 6.2) turns into a quantified risk.

3 Data

3.1 Generation: ENTSO-E BZN|ES (hourly, 2019–2024)

We download the 72 monthly files of the "Actual Generation per Production Type" dataset (16.1.B&C, area code 10YES-REE—0). The official unit is the interval-average MW (ENTSO-E, 2023): for PT60M (2019-01 to 2022-05) and PT15M (from 2022-06) the conversion to MWh per hour is either direct (PT60M) or requires a moving average (PT15M → PT60M with the mean). Coverage is 99.94% (5 hours missing on 2019-12-24 02–06h UTC, a source gap).

The dataset includes 21 *Production Types*; this work uses seven: nuclear (B14), solar (B16), wind onshore+offshore (B19+B18), reservoir hydro (B12), run-of-river (B11), pumped storage (B10) and *Fossil Gas* (B04).

3.2 Installed capacity: REData (REE)

REData publishes monthly installed capacity by technology (`api.ree.es`, no token). It is forward-filled to hourly resolution to align with ENTSO-E generation. The aggregate "Hydro" capacity (~17 GW) is not broken down into reservoir, run-of-river and pumped storage; the 35/45/20% split is used (*Section 6.4*), validated by sensitivity.

3.3 Geographic coverage

ENTSO-E BZN|ES corresponds to the peninsular system plus the Balearic Islands; it excludes the Canary Islands. REData generation-by-structure includes the Canaries. The annual difference in wind is roughly −3% (consistent with ~2 TWh/year of Canary wind), absorbed as a documented caveat. The peninsular coverage is the relevant one: SA.112483 opens the mechanism to projects "located in Spain" within the interconnected peninsular system (European Commission, 2026).

3.4 Systemic stress: MIBEL Congestion Monitor

For Axis 4 (Section 6.4) we reuse the 52 530 hours classified by Vilches (2024), of which 1 044 are Orange or Red alerts (1.99% of the total).

4 Methodology

4.1 *Naive* ELCC

$$\mathrm{ELCC}(\mathrm{tech}, N) = \frac{\overline{\mathrm{gen}}_{\mathrm{tech}}(\mathcal{T}_N)}{\bar{C}_{\mathrm{tech}}(\mathcal{T}_N)\cdot \Delta t} \tag{1}$$

where $\mathcal{T}_N$ are the N records (days in `daily`; hours in `hourly`) with the highest net demand in the period, $\bar{C}$ is the mean installed capacity over those records, and Δt is the interval duration (24h or 1h). Net demand: $D^{net} = \sum_i \mathrm{gen}_i - \mathrm{gen}_{\mathrm{wind}} - \mathrm{gen}_{\mathrm{solar}}$.

For `daily` we use $N = 100$ (top-100 days, matching the convention of ENTSO-E (2023)); for `hourly`, $N = 2\,400$ (top-2 400 hours, the same temporal volume).

4.2 The realised ELCC is the national coefficient, and equals ERAA method 12(7)(c)

The audit only bites if Eq. 1 is the object the regulator actually uses. It is. Spain's national firmness coefficient (PO, Section 7.1) is

$$\mathrm{CF}_T \;=\; \frac{1}{N_{\mathrm{hours}}}\sum_h \frac{P^{\mathrm{firm}}_{T,h}}{P^{\mathrm{inst}}_{T,\mathrm{month}}}, \tag{2}$$

averaged over the hours of *lowest coverage margin* —between 5% and 10% of the year— with P^{firm} measured per a fixed technology table and a fixed window (1-year history for thermal, 3-year for hydro). Term by term this is Eq. 1: a mean of firm-power-over-installed-capacity taken over the tightest hours (we proxy "lowest coverage margin" by "highest net demand", a choice we stress-test in §9). The same object is one of the four de-rating approaches the EU now permits —ERAA method 12(7)(c) (Agency for the Cooperation of Energy Regulators, 2026): the average of output over installed capacity across scarcity and near-scarcity hours. So for Spain the *realised* ELCC is, to a close approximation, the regulatory coefficient: the slot we audit is the real one, not a stylised stand-in.

ERAA Article 12(7) offers four approaches, of which the regulator selects one and applies it consistently (Agency for the Cooperation of Energy Regulators, 2026):

- **(a)** marginal ELCC: $\mathrm{DF} = \Delta\mathrm{EENS}_i/\Delta\mathrm{EENS}_{\mathrm{ref}}$ —the forward, marginal benchmark, not computable from realised data;
- **(b)** thermal: $\mathrm{DF} = 1 - \mathrm{FOR} - \mathrm{EPM}$, where the forced-outage rate "may reflect risks associated with fuel availability constraints";

- **(c)** average output over scarcity and near-scarcity hours, $\mathrm{DF} = \frac{1}{N}\sum \mathrm{Output}/\mathrm{Cap}$ —the realised national coefficient, = PO 7.1 = Eq. 1;
- **(d)** energy-constrained / demand response: $\mathrm{DF} = (1 - \mathrm{FOR}) \sum_e T_e C_{e,i} / \sum_e T_e$, with $C_{e,i} = \min(1, T_{\text{avail}}/T_e)$.

We therefore position our empirical ELCC as the *ex-post realised* benchmark (method (c)/PO 7.1), complementary to the forward marginal benchmark (method (a)) that the EU will publish but that realised data cannot recover.

4.3 Marginal ELCC

Following Mills and Wiser (2013) it is approximated by a finite difference:

$$\text{marginal\_ELCC(tech)} = \frac{\mathrm{ELCC}(C + \Delta C) - \mathrm{ELCC}(C)}{\Delta C} \tag{3}$$

with $\Delta C = 1\,\mathrm{GW}$. The operational assumption is that generation in peak hours does not increase when capacity is added (dispatch saturation); this provides an upper bound on the dilution from additional capacity. By construction, all marginals are negative.

4.4 Market regimes

The period is segmented into three regimes:

- *Pre-crisis*: 2019-01-01 to 2021-12-31.
- *Iberian Exception*: 2022-01-01 to 2023-12-31 (gas adjustment mechanism, RDL 10/2022).
- *Post-exception*: 2024-01-01 onwards (return to the ordinary regime with high renewable penetration).

The *regulatory bias* of a technology is defined as the difference between its ELCC over the full period and its ELCC excluding the Iberian Exception. Operationally, it is the fictitious firmness premium the 2026/2027 CM would "buy" if calibrated on the raw history.

5 Results

5.1 ELCC by technology: *daily* vs *hourly*

Table 1 summarises the ELCC obtained with both granularities, restricted to the period without the Iberian Exception. For nuclear, solar and wind the values converge ($|\Delta| < 0.01$), confirming methodological robustness. The significant differences appear in hydro and gas —precisely the technologies on which the 2026/2027 CM calibration risk falls.

5.2 Finding: disaggregated hydro

Table 1 contains the paper's main finding. The "Hydro = 0.22" aggregate of the daily approach is an opaque average of three very different technologies:

- **Reservoir** (~35% capacity): ELCC = 0.728. Almost as firm as nuclear.
- **Pumped storage** (~20% capacity): ELCC = 0.642. Dispatchable, optimised for peak hours.
- **Run-of-river** (~45% capacity): ELCC = 0.139. Low, as expected (non-dispatchable).

Table 1: ELCC by technology, *daily* (REData) vs *hourly* (ENTSO-E), without the Iberian Exception, N equivalent to 100 days.

Technology	ELCC *daily*	ELCC *hourly*	Δ
Nuclear	0.927	0.919	−0.008
Solar (PV+T)	0.126	0.126	+0.000
Wind	0.152	0.145	−0.007
Total hydro	0.222	0.334	+0.112
Reservoir hydro	—	0.728	new
Run-of-river hydro	—	0.139	new
Pumped hydro	—	0.642	new
CCGT (*daily*) / Fossil Gas (*hourly*)	0.372	0.457	+0.085

A note on reconciliation: the aggregate (0.33) is *not* the capacity-weighted mean of the three components, which is ≈0.45. The gap (≈0.11) is not an error but a definitional artefact that reinforces the point. The aggregate "total hydro" series is *net* of pumped-storage consumption (the charging load), whereas the disaggregated pumped factor (0.64) reflects gross *turbining*, which is the adequacy-relevant quantity —a pumped plant can turbine at full power in a scarcity hour irrespective of when it charged. Netting the pumping load is exactly what makes an aggregate hydro coefficient understate storage's firm contribution; the disaggregation removes that netting. The economic interpretation of the dispersion —adverse selection in the auction— is developed in Axis 1 (Section 6.1). Figure 1 shows ELCC by threshold N for all hourly technologies.

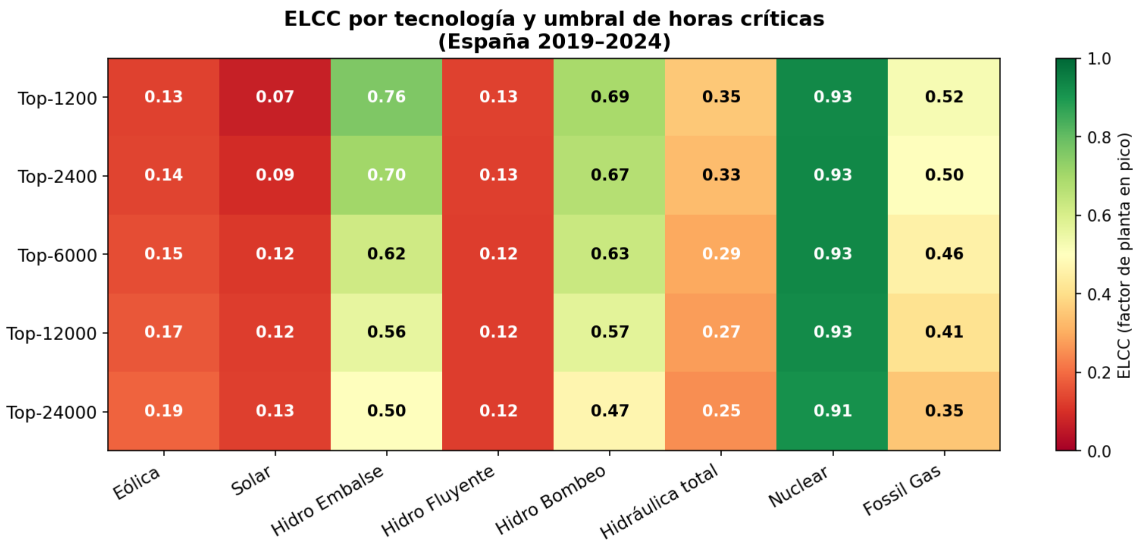

Figure 1: Hourly ELCC by technology and threshold N. The "Reservoir hydro" column is virtually flat around 0.5–0.75 across the various N, indicating high structural firmness.

5.3 Regulatory bias: the Iberian Exception

Table 2 compares ELCC with and without the Iberian Exception. For strict CCGT (*daily*, REData) the bias is +0.072 (+19% relative to 0.372). For total Fossil Gas (*hourly*, ENTSO-E, includes gas cogeneration) the bias is smaller, +0.041, because the intervention mainly affected the merit-order CCGT and not the non-intervened industrial cogeneration. The calibration implication is developed in Axis 3 (Section 6.3).

Table 2: Regulatory bias by technology (bias = $\text{ELCC}_{\text{2019-24}} - \text{ELCC}_{\text{without Iberian Exception}}$).

Technology	Bias *daily* (REData)	Bias *hourly* (ENTSO-E)
Nuclear	+0.003	+0.011
Wind	−0.024	−0.009
Solar	+0.014	−0.034
Total hydro	−0.025	−0.005
Reservoir	—	−0.024
Run-of-river	—	−0.010
Pumped	—	+0.027
CCGT / Fossil Gas	**+0.072**	**+0.041**

5.4 Marginal ELCC

Table 3 reports the marginals for $\Delta C = 1\,\text{GW}$ in *hourly*. All are negative (saturation). The most pronounced correspond to technologies with the smallest base installed capacity (pumped storage ~3.4 GW, reservoir ~6 GW, nuclear ~7 GW), where an additional GW represents a proportionally large expansion.

Table 3: Marginal ELCC, $\Delta C = 1\,\text{GW}$, *hourly*, without the Iberian Exception.

Technology	Base ELCC	ELCC +1 GW	Marginal	Base cap. (GW)
Wind	0.145	0.140	−0.0053	28.3
Solar	0.126	0.118	−0.0082	18.4
Run-of-river	0.139	0.123	−0.0160	7.7
Pumped storage	0.642	0.497	**−0.1453**	3.4
Reservoir	0.728	0.624	**−0.1042**	6.0
Nuclear	0.919	0.806	−0.1132	7.1
Fossil Gas	0.457	0.441	−0.0161	27.4

5.5 Sensitivity to HYDRO_CAP_RATIOS

REData's aggregate hydro capacity (~17 GW) is split into three sub-technologies with approximate ratios (35/45/20%) based on public REE statistics. Table 4 shows ELCC under five alternative scenarios with the aggregate capacity held fixed.

Table 4: Sensitivity of sub-hydro ELCC to HYDRO_CAP_RATIOS (top-2 400 hours, without the Iberian Exception).

Scenario	Reservoir / RoR / Pumped	Reservoir	RoR	Pumped
Base	35 / 45 / 20	**0.728**	**0.139**	**0.642**
Conservative-reservoir	30 / 50 / 20	0.849	0.125	0.642
Generous-reservoir	40 / 40 / 20	0.637	0.156	0.642
Conservative-pumped	35 / 50 / 15	0.728	0.125	0.856
Generous-pumped	35 / 40 / 25	0.728	0.156	0.514
Observed range		**[0.64, 0.85]**	**[0.13, 0.16]**	**[0.51, 0.86]**

Robustness of the finding: in the most adverse scenario for reservoir (*Generous-reservoir*, 40% cap), ELCC remains $0.637 > 0.60$. In the worst case for pumped storage (*Generous-pumped*, 25% cap), ELCC falls to 0.514, still clearly firm. The qualitative conclusion “reservoir and pumped storage are firm technologies” holds across all five scenarios —the adverse-selection risk of Axis 1 does not depend on the exact split.

5.6 Comparison with European benchmarks

Table 5 compares Spain with the UK, Ireland and Poland. The peninsular solar ELCC (∼0.13 *hourly*) exceeds the UK and Irish values by 4–13×. The reason is Spain’s bimodal demand profile: critical hours are not only winter-nocturnal but also summer-diurnal (cooling under high irradiance). Figure 2 quantifies this bimodality, the basis of Axis 2 (Section 6.2).

Table 5: Spain ELCC (*hourly*, without the Iberian Exception) vs published European benchmarks.

Country / System	Wind	Solar	Nuclear	Hydro
Spain (this work)	**0.15**	**0.13**	**0.92**	**0.33**†
UK (National Grid ESO, 2024)	0.08	0.03	0.92	0.15
Ireland (EirGrid and SONI, 2023)	0.10	0.01	—	0.20
Poland (PSE, 2023)	0.08	0.02	—	0.10

† Total hydro (Reservoir 0.73, RoR 0.14, Pumped 0.64).

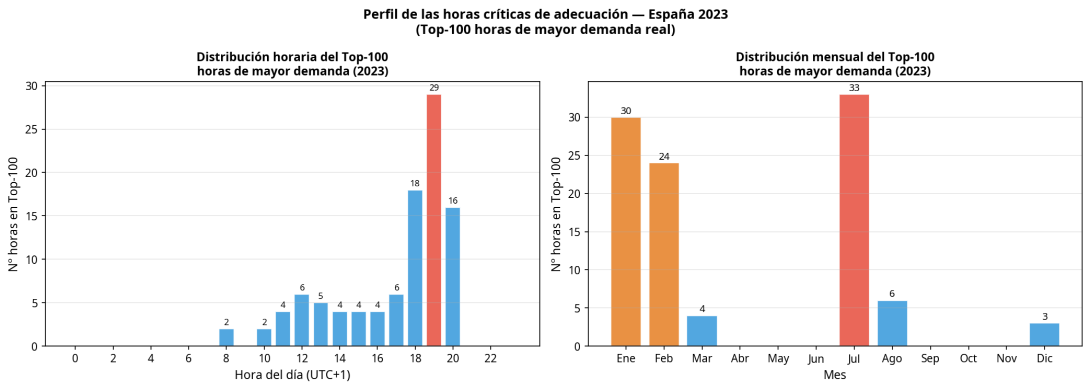

Figure 2: Hourly and monthly profile of the top-100 highest-demand hours (Spain 2023). Double peak: winter (18–20h, Jan/Dec) and summer (12–17h, Jul/Aug). ~50% of the critical hours have sun available.

5.7 Systemic stress vs adequacy: cross-check with MIBEL

The 1 044 O/R hours classified by the MIBEL Congestion Monitor (Vilches, 2024) overlap with the top-N highest-net-demand hours only marginally: a Jaccard index between 0.05 (top-50) and 0.10 (top-1 000). This confirms that the stress detected by MIBEL —structural tension on the ES-FR interconnector and extreme prices— is economic in nature, not physical adequacy. Figure 3 visualises this decoupling, the core of Axis 4 (Section 6.4).

6 Design audit: four calibration risks of the 2026/2027 CM

This section translates the results of Section 5 into design risks for the mechanism approved in SA.112483. Each axis follows the same structure: the *link* to the auction mechanics, the *argument* on the distortion introduced by shortcut calibration, and the *recommendation* for correct calibration.

Tabla 6.1: Solapamiento entre definiciones de horas críticas

N (top net demand)	Hours O/R alerts	Hours top-N	Overlap	Jaccard
50.0	1044.0	50.0	18.0	0.051
100.0	1044.0	100.0	29.0	0.074
250.0	1044.0	250.0	40.0	0.075
500.0	1044.0	500.0	64.0	0.084
1000.0	1044.0	1000.0	117.0	0.097

Figure 3: Overlap between MIBEL O/R alerts and the top-N highest-net-demand hours. Jaccard ≤ 0.10 at every threshold: economic stress is not a proxy for physical insufficiency.

6.1 Axis 1 — Aggregated hydro: adverse selection and under-remuneration of storage

Link. The auction awards a price per firm MW to each plant. The *derating factor* sets how many firm MW each unit accredits per MW installed.

Argument. If the regulator indexes the *derating factors* on "total hydro" (ELCC = 0.22), the mechanism is born economically broken. It would be *under-remunerating* reservoir (0.73) and pumped storage (0.64) by a factor close to 3×, and *over-remunerating* run-of-river (0.14) by almost double. The result is a classic *adverse-selection* problem: run-of-river —which, depending on instantaneous flow, cannot guarantee power in an hour of physical stress— receives a firmness premium it does not deserve, while reservoir and pumped storage —the hydro storage that does deliver power in scarcity— are underpaid. The long-term investment signal for storage, which SA.112483 states it wants to foster to reach the national flexibility target (European Commission, 2026), would be destroyed in the very first auction parameter. The sensitivity of Section 5.5 confirms that reservoir stays firm ($\in [0.64, 0.85]$) and pumped storage too ($\in [0.51, 0.86]$) across the five scenarios: the risk is not an artifact of the assumed capacity split.

A subtler screening flaw survives even when the regulator *does* disaggregate, as PO 7.1 in fact does (reservoir and run-of-river are separate table rows). The PO measures hydro firm power by *producción media horaria* —mean production— not *potencia disponible*, because hydro is deemed conditioned by a primary resource (water). For run-of-river that is right; but a *reservoir* is dispatchable within its stored energy, and its mean production in the critical hours reflects how it was *operated* (often holding water for later), not the power it could *deliver* if called. Screening a dispatchable reservoir by average production therefore under-reveals its type: the empirical 0.73 is the delivered-under-stress contribution the production channel hides. The aggregate "0.22" is the cruder form of the same error —the data problem this paper solves— while the production-versus-availability mis-measurement is the screening form that persists after disaggregation.

Recommendation. Disaggregate hydro into differentiated auctions or, at a minimum, calibrate *derating factors* specific to each production-unit code (ESIOS), with values around 0.73 (reservoir), 0.64 (pumped) and 0.14 (run-of-river). This is the only way to mitigate adverse selection and preserve the storage investment signal that CISAF itself prioritises.

6.2 Axis 2 — Bimodal profile: solar *missing money*

Link. The solar *derating factor* determines what fraction of solar *missing money* the plant recovers via capacity revenue.

Argument. The CM design in northern Europe (UK, Poland, Ireland) assumes a virtually nil solar adequacy value (1–3%) because there critical hours occur strictly on winter nights. But the Spanish profile is *bimodal* (Figure 2): the top of critical hours includes a massive summer peak (July/August, 12–17h) driven by cooling. With $\text{ELCC}_{\text{solar}} = 0.13$, peninsular PV physically coincides with ~50% of the system's critical adequacy hours. If the 2026/2027 CM blindly imports the National Grid or PSE parameters, it would penalise solar *missing money* by ~80% (from 0.13 to ∼0.03), a de facto discrimination that the non-discrimination criterion of SA.112483 (European Commission, 2026) does not allow. In screening terms the *window* is the auditable parameter: which hours count as "critical" (the lowest-coverage-margin set) is precisely what makes the solar type reveal as ≈0.13 or ≈0.03. The window is a channel-design choice, not a fact of nature.

A second, structural gap compounds this. PO 7.1 calibrates the coefficient on *realised history*, in which Spain's realised LOLE is ≈0 —the critical hours are relatively-tight, not genuinely scarce— yet the coefficient is then applied to a *forward* world whose national adequacy assessment projects LOLE ≈6.26 h (the very gap that justifies the mechanism against the 1.5 h standard). The slot is thus calibrated on a structurally different period from the one it will govern: history-without-scarcity used to screen a future-with-scarcity. Solar's delivered value in the tight summer hours of the historical window need not equal its value in a genuinely scarce forward system, and the static coefficient cannot tell the difference.

Recommendation. Calibrate the solar *derating factor* on the domestic bimodal profile (∼0.13), not on imported benchmarks, and make the critical-hour window explicit and forward-consistent rather than a silent historical artefact.

6.3 Axis 3 — Gas: a deliberate availability choice and its fuel-blind residual

Link. Combined-cycle plants (CCGT) will be a key back-up resource in the first auctions; their *derating factor* sets the firm MW they accredit.

Argument. Section 7.1 of the PO fixes a classification rule: measure firm power by *potencia disponible* (declared technical availability) *unless* a technology's contribution is conditioned by the availability of a primary energy resource, in which case measure it by *producción media horaria* (mean hourly production). The accompanying table assigns CCGT, nuclear and coal to *potencia disponible* —net capacity minus communicated unavailabilities under PO 3.6, so that an idle but non-faulted plant still counts as available— over a one-year window. Yet a CCGT's contribution *is* conditioned by a primary energy resource: natural gas. By the PO's own rule the gas factor should then be measured by production. We read the table's choice not as a regulatory error but as a deliberate and, for the most part, sound one: measuring thermal by availability is precisely what *insulates* the coefficient from the production-side distortions of subsidy episodes —as the decoupling we measure below confirms, the Iberian Exception moved gas dispatch, not declared availability. The choice is consistent, however, only under an unstated assumption that fossil-fuel supply is always available and non-binding —a *fuel-blindness* that the 2022 European gas crisis falsified, and which is the one residual the availability route leaves.

Put plainly, the rule and its table point in different directions for gas, and each route leaves a distinct gap. *(i) The table* (availability): the coefficient is blind to fuel-supply risk; it credits the plant's technical ability to start and ignores that gas delivers only if fuel is physically there. *(ii) The rule* (production, since fuel is a primary resource): the coefficient is contaminated by the Iberian Exception, whose gas subsidy inflated CCGT dispatch —the +0.072 (+19%) we measure on production (Table 2). The PO cannot satisfy both readings simultaneously; the gas coefficient is mis-specified under either.

It is fuel, not price, that is the durable problem. In a true scarcity event the spot price approaches the value of lost load, so gas is economic to run regardless of any subsidy; the apparent price-conditionality of CCGT output is an artifact of measuring firmness over a non-scarcity proxy window (the peninsular system's realised LOLE is near zero), not a property of scarcity itself. What binds in scarcity is fuel: a CCGT's adequacy contribution depends on gas being deliverable, and a gas-supply crunch —the most plausible adequacy threat for an import-dependent peninsula— is precisely the state in which the fleet under-delivers together.

Here the national method is weakest. The EU benchmark anticipates exactly this: under the ERAA methodology, Art. 12(7)(b), the thermal derating factor is 1 − FOR − EPM and the forced-outage rate "may reflect risks associated with fuel availability constraints". The PO's *potencia disponible* (PO 3.6) captures technical unavailability only, not fuel-supply risk —it is weakest in the exact dimension that 2022 exposed.

Anticipated objection, and why it fails. One could reply that the PO credits availability ex ante but penalises non-delivery ex post: the availability obligation carries a charge (PO, Section 13.2.1: $\text{OPINCSCAP}_1 = P_{inc} \cdot \text{Price} \cdot K_1$, with $K_1 = 1.2$) enforced through availability tests, so any fuel-conditionality is settled after the fact. But penalties enforce *individual* delivery; they cannot price *correlated* fuel risk. A system-wide gas crunch tumbles the whole CCGT fleet at once: every plant pays its penalty and the lights still go out. Over-crediting gas firmness is therefore an *adequacy* failure, not a settlement one —one cannot penalise one's way out of a blackout.

The decoupling, measured. The +0.072 is computed on *production* (ENTSO-E generation). We can now show directly that it does *not* pass through the availability channel the PO uses. Pulling the CCGT declared-available-power series (ESIOS indicator 477, national, 2019–2024) and restricting to the 5% highest-net-demand hours, available power is essentially flat across regimes —≈21.2 GW pre-crisis, ≈20.7 GW during the Iberian Exception, ≈19.4 GW post— while CCGT *production* in those same hours jumps from ≈12.9 to ≈14.8 GW during the Exception and reverts to ≈11.8 GW after (utilisation 61% → 72% → 61%). The Exception was a dispatch shock, not an availability shock: the contamination lives in the production channel and bites with full force on the technologies the PO measures by production (reservoir and run-of-river hydro, solar, wind; Axes 1–2). For *gas*, measured by availability, the production contamination does not transmit —but the same data expose the other horn: in normal regimes ≈21 GW is declared available while only ≈12 GW is delivered in the scarcity hours (61% utilisation), so the availability coefficient over-credits delivery by the fuel/economic conditioning it ignores. (The availability series is strict CCGT while our production series is the broader "Fossil Gas"; the utilisation level is therefore approximate, but the flat-availability-versus-spiking-production decoupling is robust to the definitional mismatch.)

Recommendation. Condition the gas *derating factor* on fuel-supply risk, as ERAA Art. 12(7)(b) permits through the forced-outage rate, rather than on *potencia disponible* alone; and anchor the firmness window on genuine scarcity states so that neither the Exception's dispatch inflation nor a non-scarcity proxy distorts the coefficient. A gas factor blind to fuel risk over-credits the most conditional resource in the fleet —an adverse selection the screening auction then acts on.

6.4 Axis 4 — Jaccard decoupling: the CM is not a price insurance

Link. There is a design tension over whether the CM should measure its stress —and trigger its non-delivery penalties— from the *spot* price or from physical variables.

Argument. The cross-check with the MIBEL Congestion Monitor is conclusive: a Jaccard index ≤ 0.10 between economic-stress alerts and the top of net demand (Figure 3) shows that price stress —typically caused by congestion rents on the interconnector with France or gas price spikes— *does not correlate* with the physical insufficiency of internal generation. A CM that activated penalties on price criteria would be remunerating and penalising the wrong event.

Recommendation. Validate the mechanism's separate design: the CM should be a complement to the *spot* market, indexing its non-delivery penalties strictly to hours of *physical* adequacy stress, not to spikes of *day-ahead* financial volatility. Activating penalties on price would distort plants' risk hedging and inflate bid premiums —raising the cost of the mechanism for the consumer, against the Commission's first criterion (European Commission, 2026).

6.5 Marginal ELCC: do not over-remunerate redundant capacity

Cutting across the four axes, Table 3 shows that all marginals are negative (dilution from saturation). Auctions should incorporate marginal-cost calculation methodologies so as not to over-remunerate additions in already saturated technologies that bring no adequacy value at the margin.

7 The headline distortion: the 2027/2029 décalage and a 15-year lock-in

The four axes establish that the national slot mis-screens firmness. Their *consequence* is a timing failure, and it is the paper's strongest claim because it survives the objection that the EU method is well designed: a good slot that arrives after the contracts are signed disciplines nothing.

Spain's first auctions, expected in Q1 2027, set de-rating through the national historical method (PO 7.1), which —as §4 shows— is the paper's own Eq. 1 and ERAA method 12(7)(c). The standardized EU de-rating parameters, by contrast, phase in only gradually: under ACER Decision 04/2026 (Agency for the Cooperation of Energy Regulators, 2026) (Annex I, "Implementation of the methodology", and recital 66) the amended methodology applies from the ERAA 2027 edition (ERAA 2026 still runs ACER Decision 24/2020 (Agency for the Cooperation of Energy Regulators, 2020)), with the CM-related parameters delivered as a proof-of-concept in ERAA 2026, a single, *simplified* target year in ERAA 2027, two target years in ERAA 2028, and full coverage of all target years only from ERAA 2029. So at the moment the Q1-2027 contracts are signed, no full standardized EU de-rating factors for Spain exist —at best a partial, simplified, single-target-year output.

The horizons turn this gap into a decade-long commitment. Under the notified design (draft order, SA.112483; status flagged) new-investment contracts run up to *fifteen* years, existing plant is re-contracted annually, and new demand 1–10 years. Because the $\leq$550 g CO_2/kWh eligibility excludes new gas build, the 15-year contracts accrue to *new renewables, storage and demand* —exactly the resources Axis 1 (storage) and Axis 2 (solar) show the national slot under-credits. The Axis-3 gas distortion, by contrast, sits on existing plant re-set annually, so its lock-in is short. The mis-screening is therefore locked in *precisely where the investment signal lives*: a storage or solar project that clears the 2027 auction on an under-stated coefficient carries that price for up to fifteen years, while the cleaner EU benchmark does not even reach full coverage until 2029. The empirical ELCC of §5 quantifies the magnitude of what is locked. In screening terms: the slot is mis-designed, the captive technologies self-select under it, and by the time the EU slot matures the segment is already contracted for its full horizon.

8 Penetration- and mix-blindness of the static national coefficient

A second structural limitation compounds the lock-in. ACER's amended methodology explicitly recognises that "the capacity value of energy-limited resources, such as battery storage and demand response, may vary materially with their penetration level and with the availability of complementary energy-providing resources" (ACER Decision 04/2026, recital 199), that deriving factors "under a single, implicitly assumed capacity mix may not fully capture these interaction effects" (recital 200), and accordingly amends Article 12 with a new paragraph 8 allowing ENTSO-E to compute "supplementary sets of de-rating factors reflecting alternative plausible capacity mixes", which "may also support Member States that apply *dynamic* de-rating factors" (recitals 202–203). The EU framework, in short, treats firmness as mix- and penetration-dependent.

The national PO 7.1 produces the opposite: a *single, static* historical coefficient (1-year thermal, 3-year hydro), blind to penetration and to the prevailing mix. The interaction with §7 is the killer point. A static coefficient struck in 2027 and applied to a fifteen-year contract is blind to the penetration-driven *decline* in the capacity value of solar and storage over the contract's life —the very saturation effect our marginal-ELCC estimates (Table 3, all negative) display and that ACER itself says matters. The method is structurally incapable of capturing the mix evolution a fifteen-year contract spans. We note the supply-versus-demand caveat: the average method (12(7)(c), the national choice) is consistent with summing supply-side firm capacity but is the less accurate, non-marginal branch, and the national procedure performs none of the alternative-mix analysis that Article 12(8) now permits.

9 Identification, scope and robustness

We set out what the result rests on and what it deliberately does not claim.

The metric is the realised coefficient. Our object is the *ex-post realised* ELCC —the average approach 12(7)(c), which is exactly PO 7.1 (§4.2)— not the forward marginal metric (12(7)(a)). The two are complementary: we audit the slot Spain will actually use and benchmark, rather than replace, the forward factor the EU will publish.

The critical-hour definition. PO 7.1 itself screens on a non-ENS proxy (lowest coverage margin); our top-net-demand set is the same family, and the technology ranking is invariant across top-N and coverage-margin quantiles.

The solar window is the parameter, not a weakness. Solar's coefficient moves with the window (0.13 vs. 0.03); that dependence is the finding (Axis 2), reported across windows rather than fixed silently.

The gas channel is precisely scoped. The gas result is the table-versus-rule dilemma and fuel-blindness (§6.3), with the production-versus-availability decoupling measured; the production contamination is confined to the production channel and is not attributed to declared availability.

We audit the admissible space, not the EU design. The ERAA method is sound; what we audit is the latitude it leaves —which of the four approaches, which window, which history— and the national/EU two-speed gap (§7): a sound method that lands after the 2027 contracts disciplines nothing.

Coverage bounds magnitude, not sign. The Canaries are excluded (∼3% on wind, an isolated system); the 35/45/20 hydro split is assumed but the finding survives five splits (Table 4); only 2024 is post-Exception; "Fossil Gas" $\neq$ strict CCGT, so the +0.072 uses strict CCGT. Each affects the magnitude, none the direction.

Average versus marginal. Eq. 1 is an average and an upper bound on the marginal contribution; Table 3 characterises the gap per technology.

Penetration-dependence is a result, not a gap. Firmness is mix- and penetration-dependent, which the static national coefficient cannot capture; we make this a finding (§8), on ACER's own recitals 199–203 and new Article 12(8).

Legal status, used affirmatively. The enabling provision was repealed and reinstated (Article 33(13) of Ley 24/2013 via RDL 7/2026 (Jefatura del Estado, 2026)); the operational order is draft and the SA.112483 conditions pending; the first auction is expected Q1 2027. Every implementation-dependent parameter is flagged, and the unsettled base is itself part of the path-dependence finding.

10 Limitations

10.1 Geographic coverage

ENTSO-E BZN|ES excludes the Canaries. The ~2 TWh/year of Canary wind is not reflected, producing a downward bias of ~3% in wind ELCC. As the Canaries are an isolated system with its own dispatch mechanism, this exclusion is defensible for the peninsular CM.

10.2 Fossil Gas definition

ENTSO-E "Fossil Gas" includes natural-gas-fired cogeneration, not only CCGT. This is functionally correct for measuring the system's total firm gas, but it is not 1:1 with the "Combined Cycle" category that the CM treats as a separate auction. Axis 3 therefore uses strict CCGT (REData) to quantify the bias.

10.3 Internal hydro split

REData publishes aggregate hydro installed capacity (~17 GW) without breaking down reservoir, run-of-river and pumped storage. The 35/45/20% split is assumed, based on public REE statistics. The sensitivity (Section 5.5, Table 4) confirms that the qualitative finding is robust, but precise numerical calibration would require access to ENTSO-E 14.1.A (Installed Capacity per Production Type), pending.

10.4 Plant-level granularity

We use ESIOS for the CCGT declared-availability series of §6.3 (indicator 477). What remains out of reach is generation *per unit* (rather than per production type), which would enable plant-level ELCC and the per-plant hydro *derating factors* Axis 1 recommends; that refinement is future work. The gas result of §6.3 is therefore complete at the technology level (the availability-versus-production decoupling is measured) but not yet at the unit level.

10.5 Short post-exception period

Only 2024 is available as a post-Exception observation. The findings on the post-exception regime (especially expanded solar and pumped storage) should be revisited with 2025–2026 data, ideally before the mechanism's first auction.

11 Conclusions

1. For nuclear, solar and wind, hourly granularity does not materially change ELCC relative to daily ($|\Delta| < 0.01$). The daily REData approach is methodologically valid for these technologies.

2. **Axis 1.** The "Hydro = 0.22" aggregate conceals three very different regimes (**reservoir 0.73, run-of-river 0.14, pumped 0.64**). Calibrating the CM on the aggregate generates adverse selection: it underpays storage $\sim 3\times$ and overpays run-of-river. The finding is robust to five capacity-split scenarios (reservoir $[0.64, 0.85]$, pumped $[0.51, 0.86]$).

3. **Axis 2.** The peninsular solar ELCC (0.13) is 4–13× the northern-European benchmarks because of the bimodal profile. Importing foreign parameters would penalise solar *missing money* by ~80%.

4. **Axis 3.** The gas coefficient is mis-specified under the PO's own rules: by declared availability (PO 3.6) it is blind to fuel-supply risk; by production it carries the Iberian Exception's +0.072 dispatch inflation. The durable fix is to condition gas firmness on fuel risk, as ERAA Art. 12(7)(b) permits, not to trust *potencia disponible*. The +0.072 is a production-channel result and bites directly on the production-measured technologies (hydro, solar, wind); for gas, the availability-versus-production decoupling is measured (ESIOS 477: available power flat across regimes while dispatch spikes in the Exception), confirming both that the contamination does not reach the availability coefficient and that the coefficient over-credits delivery.

5. **Axis 4.** MIBEL stress is economic (Jaccard ≤ 0.10 with physical adequacy). The CM should index penalties on physical stress, not on the *spot* price; the mechanism's separate design is consistent with this evidence.

6. **Décalage and lock-in.** The first auctions (Q1 2027) run the national slot while the standardised EU de-rating factors reach full coverage only with ERAA 2029; new-investment contracts run up to fifteen years and accrue (via the $\leq$550 g CO_2 limit) to renewables, storage and demand. The mis-screened storage/solar coefficient is therefore locked a decade before the cleaner benchmark lands —exactly where the investment signal lives— while the gas distortion, sitting on annually re-set existing plant, is short-lived.

7. **Penetration-blindness.** The static national coefficient is blind to the mix- and penetration-dependence of firmness that ACER itself recognises (Decision 04/2026, recitals 199–203; new Article 12(8)); applied to a 15-year contract it cannot see the penetration-driven decline in solar and storage capacity value the contract spans.

8. **Joint reading.** Read through the screening lens, the four facets are one adverse-selection failure: a slot that mis-reveals firmness types, priced against the value of lost load (22,879 €/MWh) over a multi-year, ≈900 € million/year stream. Getting SA.112483 right is not "computing ELCC": it is designing the slot so the firm resource self-selects in —and pricing it before, not after, the contracts that lock it for a decade are signed.

Reproducibility The full code, data and figures are available at https://github.com/cmvdata/capacity. Results are reproducible via `python main.py -freq hourly`; the unit tests (`pytest tests/`, 27/27 *passing*) cover the critical methodological decisions.

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